

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Change, Update]	179554	49.311088
1	[Git, Change, Create]	52989	14.552420
2	[Git, Change, Delete]	27984	7.685273
3	[Java, ByteCode, Member, Method]	13494	3.705870
4	[Git, Commit]	13475	3.700652
5	[Java, ByteCode, Parameter]	13293	3.650669
6	[File, Git]	11237	3.086028
7	[Git, Change, Rename]	10830	2.974253
8	[Java, ByteCode, Bound, ParameterizedType]	7279	1.999039
9	[Java, ByteCode, Bound]	7240	1.988328
10	[Java, ByteCode, Member, Field]	3609	0.991143
11	[Java, ByteCode, Bound, WildcardType]	2947	0.809337
12	[Java, Value, ByteCode, Annotation]	2931	0.804943
13	[Xml, Element]	2162	0.593752
14	[Java, ByteCode, Member, Method, Constructor]	2120	0.582218
15	[Xml, Text]	1450	0.398215
16	[Java, ByteCode, TypeVariable, Bound]	1111	0.305115
17	[Java, ByteCode, Member, Method, Lambda]	968	0.265843
18	[Type, File, Java, ByteCode, ResolvedDuplicate...	888	0.243872
19	[Type, File, Java, ByteCode, Class]	846	0.232338
20	[Json, Key]	702	0.192791
21	[Value, Json, Scalar]	685	0.188122
22	[Java, Value, ByteCode, Primitive]	672	0.184552
23	[Type, File, Java, ByteCode, JavaType]	649	0.178235
24	[Java, GenericDeclaration, ByteCode, Member, M...	578	0.158737
25	[Type, File, Java, ByteCode, ExternalType]	411	0.112873
26	[Git, Change, Copy]	347	0.095297
27	[Author, Git, Person]	299	0.082115
28	[Value, Array]	287	0.078819
29	[Committer, Git, Person]	252	0.069207

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

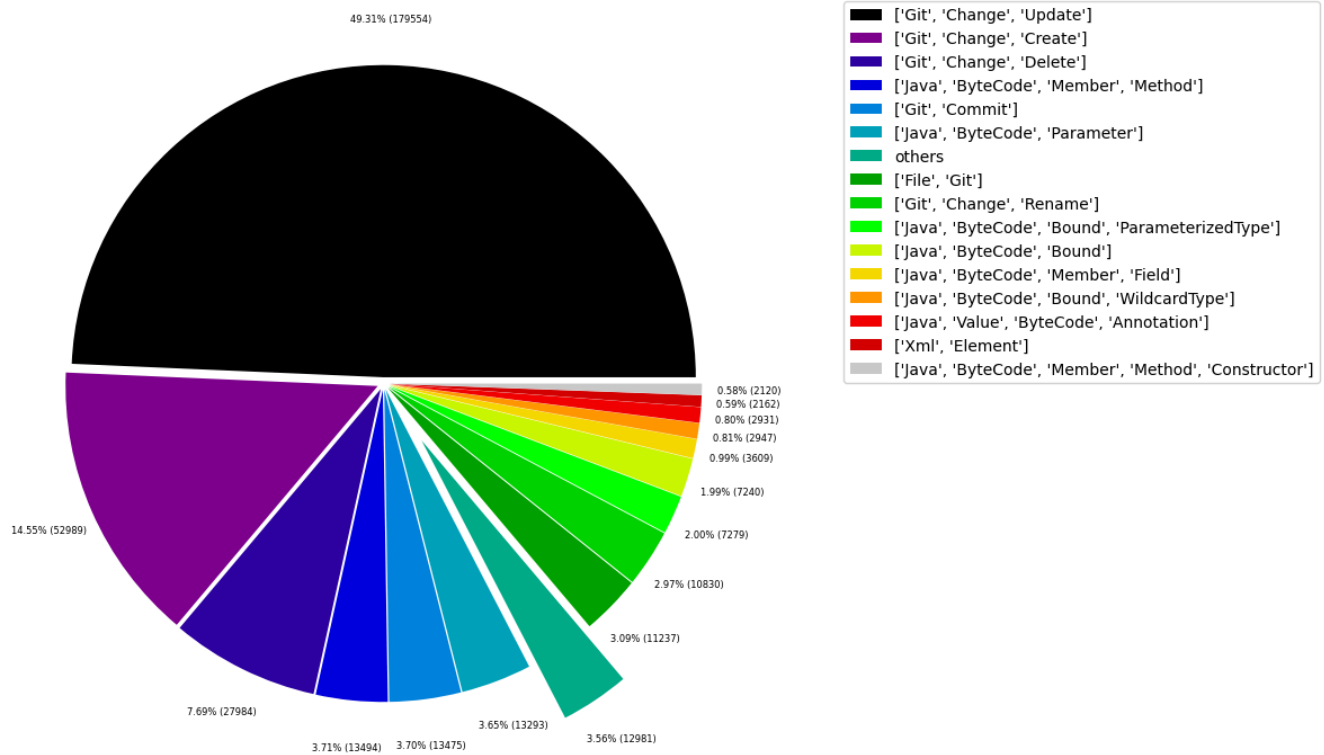


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000275
1	[Package, File, Json, NPM]	1	0.000275
2	[Repository, File, Git]	1	0.000275
3	[File, TS, Scan]	1	0.000275
4	[File, Json]	2	0.000549
5	[File]	3	0.000824
6	[Java, GenericDeclaration, ByteCode, Member, M...	4	0.001099
7	[Maven, Exclusion]	5	0.001373
8	[Value, Array, Json]	6	0.001648
9	[Dependency, NPM]	7	0.001922
10	[Type, File, Java, ByteCode, Void]	9	0.002472
11	[File, Maven, Xml, Pom, Document]	9	0.002472
12	[Java, ManifestSection]	9	0.002472
13	[File, Java, Manifest]	9	0.002472
14	[File, Artifact, Jar, Archive, Zip, Java]	9	0.002472
15	[File, Java, ServiceLoader]	10	0.002746
16	[File, Java, Properties]	12	0.003296
17	[Maven, ExecutionGoal]	16	0.004394
18	[Maven, PluginExecution]	16	0.004394
19	[Xml, Attribute]	18	0.004943
20	[jQAssistant, Rule, Concept]	19	0.005218
21	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.005767
22	[Maven, Configuration]	21	0.005767
23	[Maven, Plugin]	21	0.005767
24	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.006042
25	[Type, File, Java, ByteCode, Enum]	28	0.007690
26	[Type, File, Java, ByteCode, PrimitiveType]	30	0.008239
27	[Xml, Namespace]	36	0.009887
28	[Type, File, Java, ByteCode, Annotation]	44	0.012084
29	[Git, Branch]	48	0.013182

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

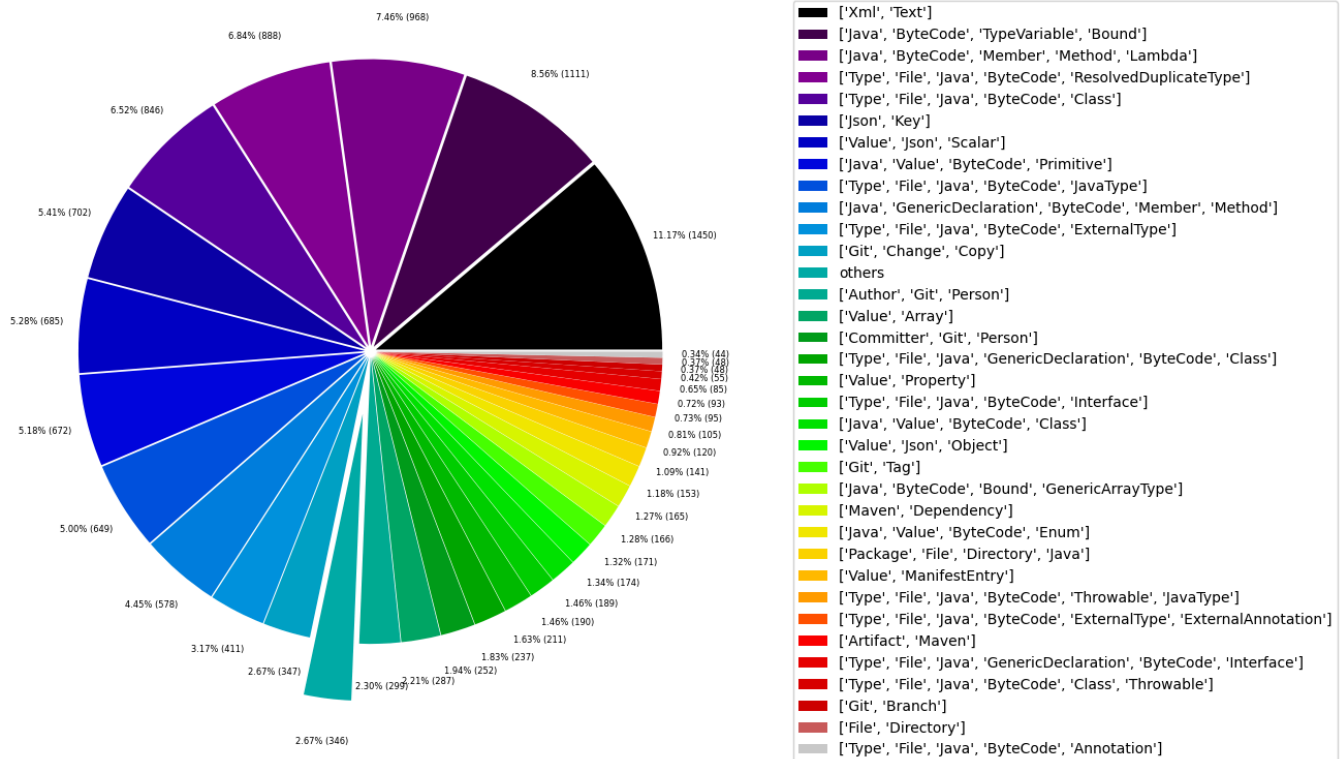


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

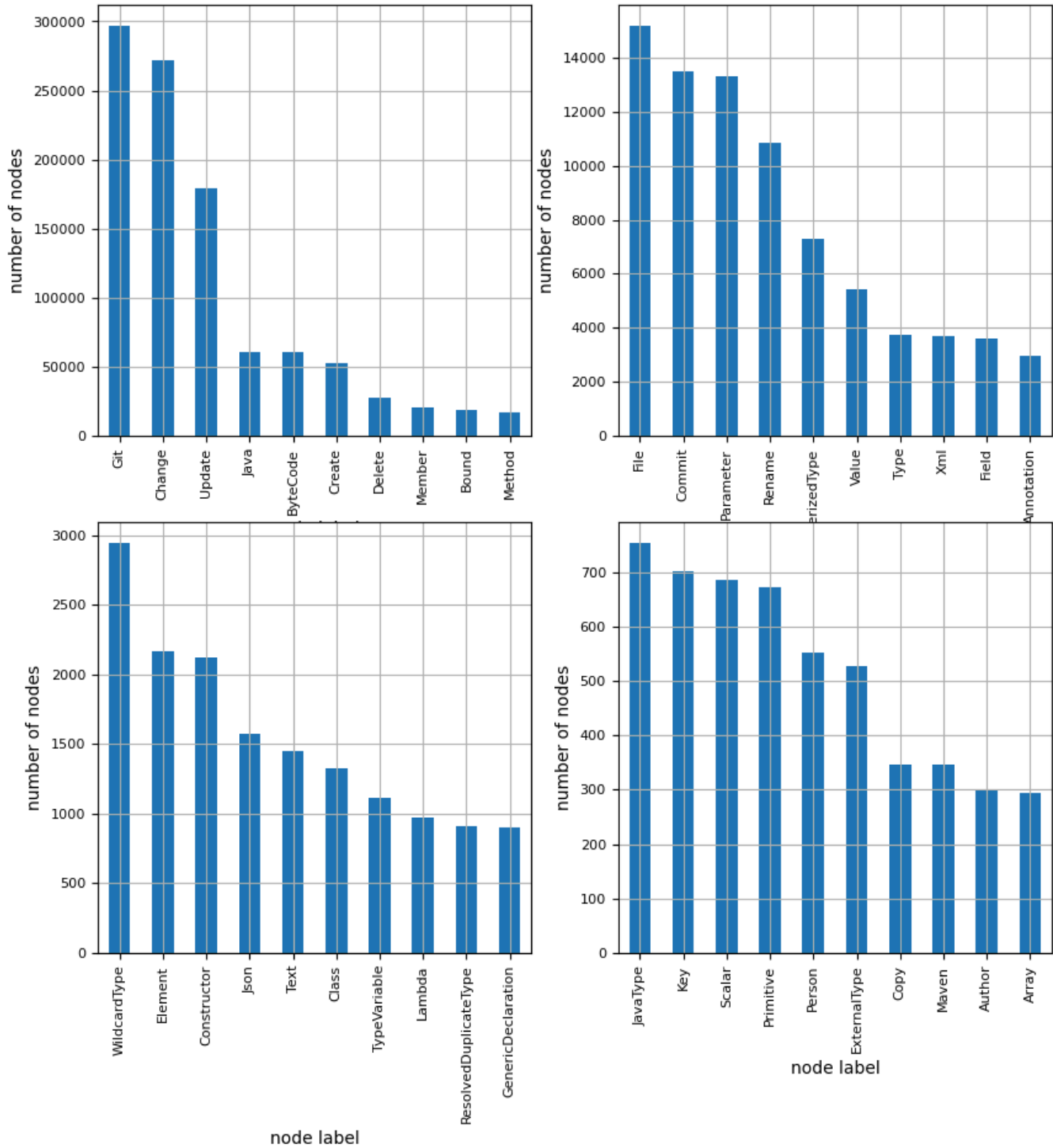
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	297187	81.616752
1	Change	271704	74.618332
2	Update	179554	49.311088
3	Java	60659	16.658840
4	ByteCode	60469	16.606660
5	Create	52989	14.552420
6	Delete	27984	7.685273
7	Member	20773	5.704909
8	Bound	18743	5.147408
9	Method	17164	4.713766
10	File	15198	4.173841
11	Commit	13475	3.700652
12	Parameter	13293	3.650669
13	Rename	10830	2.974253
14	ParameterizedType	7279	1.999039
15	Value	5428	1.490697
16	Type	3715	1.020254
17	Xml	3675	1.009269
18	Field	3609	0.991143
19	Annotation	2975	0.817027
20	WildcardType	2947	0.809337
21	Element	2162	0.593752
22	Constructor	2124	0.583316
23	Json	1570	0.431171
24	Text	1450	0.398215
25	Class	1327	0.364435
26	TypeVariable	1111	0.305115
27	Lambda	968	0.265843
28	ResolvedDuplicateType	910	0.249914
29	GenericDeclaration	904	0.248266
30	JavaType	754	0.207072
31	Key	702	0.192791
32	Scalar	685	0.188122
33	Primitive	672	0.184552
34	Person	551	0.151322
35	ExternalType	527	0.144731
36	Copy	347	0.095297
37	Maven	346	0.095022
38	Author	299	0.082115
39	Array	293	0.080467

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 1130715

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	271704	24.029397
1	MODIFIES	271704	24.029397
2	UPDATES	179554	15.879687
3	CREATES	64166	5.674816
4	DELETES	38814	3.432695
5	INVOKES	36776	3.252455
6	COMMITTED	26950	2.383448
7	DEPENDS_ON	22492	1.989184
8	OF_TYPE	21920	1.938596
9	DECLARES	21234	1.877927
10	OF_RAW_TYPE	17353	1.534693
11	HAS_PARENT	16276	1.439443
12	HAS	14429	1.276095
13	HAS_COMMIT	13475	1.191724
14	RETURNS	12844	1.135918
15	HAS_FILE	11237	0.993796
16	RENAMES	10830	0.957801
17	READS	9378	0.829387
18	HAS_ACTUAL_TYPE_ARGUMENT	8407	0.743512
19	HAS_NEW_NAME	6350	0.561592
20	OF_GENERIC_TYPE	5996	0.530284
21	RESOLVES_TO	5270	0.466077
22	SIMILAR	4078	0.360657
23	WRITES	3944	0.348806
24	CONTAINS	3936	0.348098
25	RETURNS_GENERIC	3593	0.317764
26	ANNOTATED_BY	2919	0.258155
27	REQUIRES	2230	0.197220
28	HAS_FIRST_CHILD	2162	0.191206
29	HAS_LAST_CHILD	2162	0.191206

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

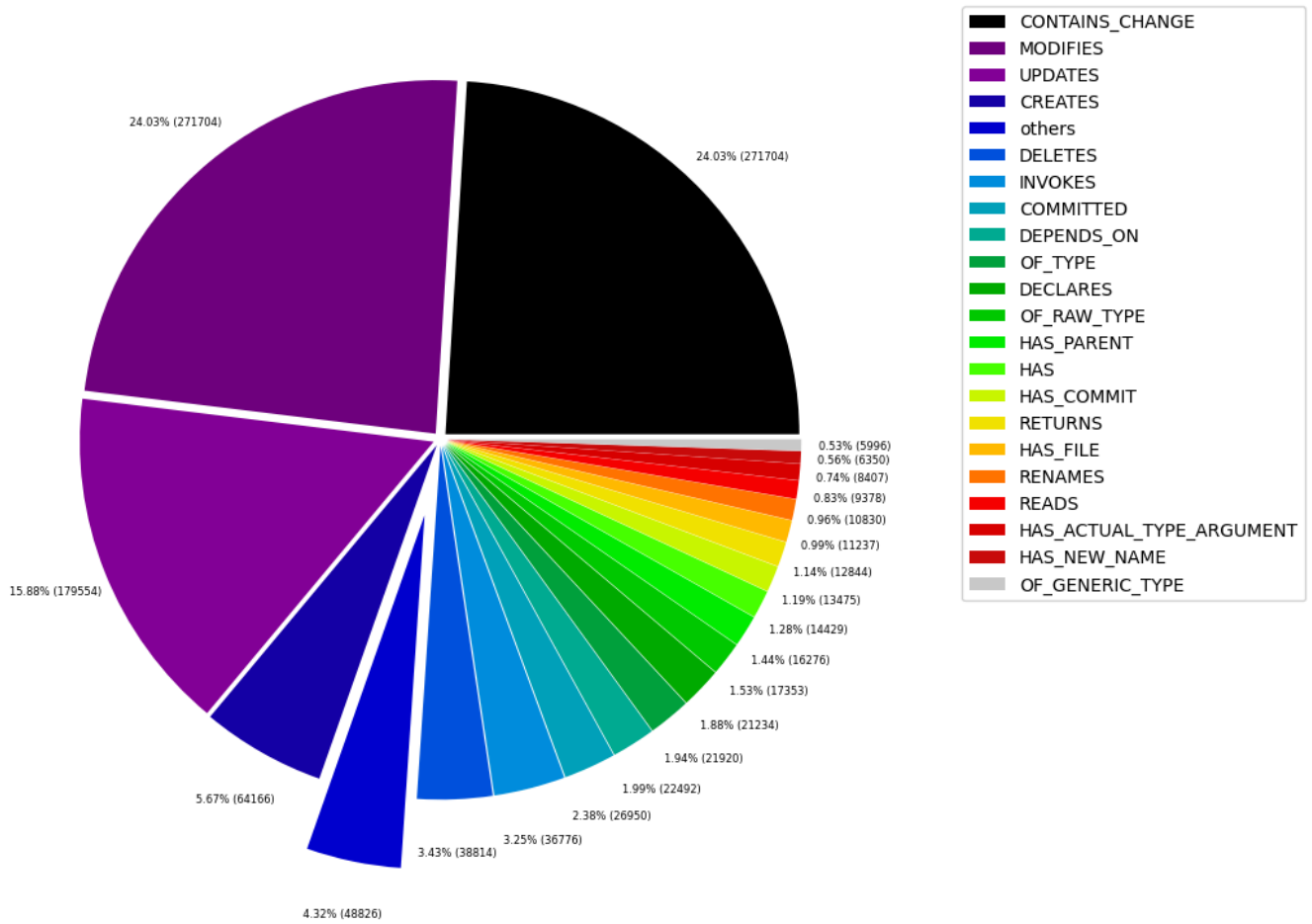


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

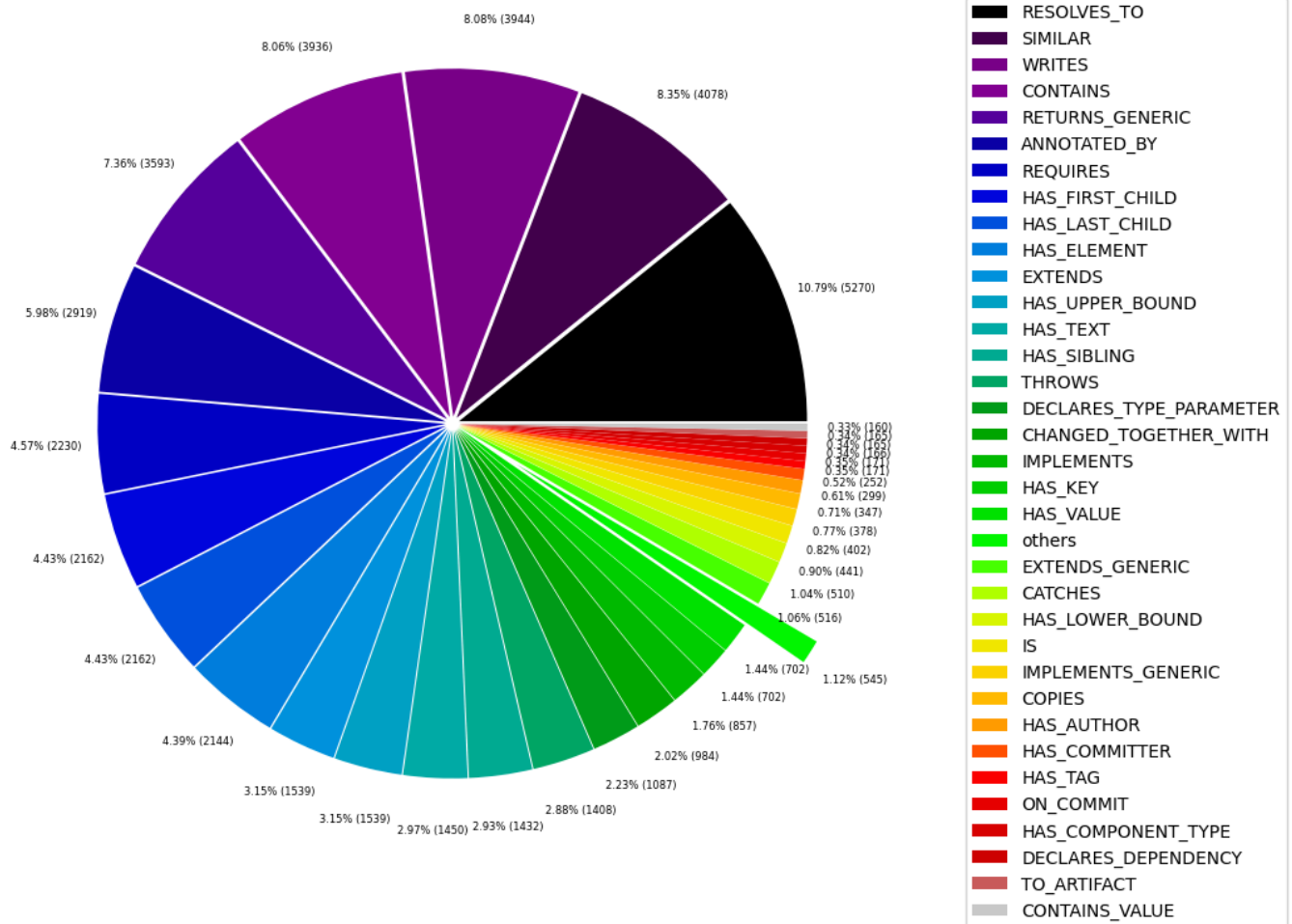
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000088
1	THROWS_GENERIC	5	0.000442
2	EXCLUDES	5	0.000442
3	DECLARES_DEV_DEPENDENCY	7	0.000619
4	DESCRIBES	9	0.000796
5	HAS_ROOT_ELEMENT	12	0.001061
6	HAS_GOAL	16	0.001415
7	HAS_EXECUTION	16	0.001415
8	OF_NAMESPACE	18	0.001592
9	HAS_ATTRIBUTE	18	0.001592
10	INCLUDES_CONCEPT	19	0.001680
11	IS_ARTIFACT	21	0.001857
12	HAS_CONFIGURATION	21	0.001857
13	USES_PLUGIN	21	0.001857
14	REQUIRES_TYPE_PARAMETER	24	0.002123
15	REQUIRES_CONCEPT	28	0.002476
16	DECLARES_NAMESPACE	36	0.003184
17	HAS_DEFAULT	39	0.003449
18	HAS_BRANCH	48	0.004245
19	HAS_HEAD	49	0.004334
20	COPY_OF	132	0.011674
21	CONTAINS_VALUE	160	0.014150
22	DECLARES_DEPENDENCY	165	0.014593
23	TO_ARTIFACT	165	0.014593
24	HAS_COMPONENT_TYPE	166	0.014681
25	ON_COMMIT	171	0.015123
26	HAS_TAG	171	0.015123
27	HAS_COMMITTER	252	0.022287
28	HAS_AUTHOR	299	0.026443
29	COPIES	347	0.030689

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Update]	179554	13475	
1	[Git, Change, Update]	UPDATES	[File, Git]	179554	179554	
2	[Git, Change, Update]	MODIFIES	[File, Git]	179554	179554	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	52989	13475	
4	[Git, Change, Create]	CREATES	[File, Git]	52989	52989	
5	[Git, Change, Create]	MODIFIES	[File, Git]	52989	52989	
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	27984	13475	
7	[Git, Change, Delete]	MODIFIES	[File, Git]	27984	27984	
8	[Git, Change, Delete]	DELETES	[File, Git]	27984	27984	
9	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	22590	13404	
10	[Git, Commit]	HAS_PARENT	[Git, Commit]	16267	13475	
11	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	13475	1	
12	[Committer, Git, Person]	COMMITTED	[Git, Commit]	13475	252	
13	[Author, Git, Person]	COMMITTED	[Git, Commit]	13475	299	
14	[Repository, File, Git]	HAS_FILE	[File, Git]	11237	1	
15	[Git, Change, Rename]	RENAMES	[File, Git]	10830	10830	
16	[Git, Change, Rename]	DELETES	[File, Git]	10830	10830	
17	[Git, Change, Rename]	CREATES	[File, Git]	10830	10830	
18	[Git, Change, Rename]	MODIFIES	[File, Git]	10830	10830	
19	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	10830	13475	
20	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	8517	13404	
21	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	8446	13404	
22	[File, Git]	HAS_NEW_NAME	[File, Git]	6350	11237	
23	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	6156	13293	
24	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3465	7240	
25	[Java, ByteCode, Bound, ParameterizedType]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3110	7279	
26	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2947	7279	
27	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, Bound, ParameterizedType]	2695	13293	
28	[Java, ByteCode, Member, Method, Constructor]	WRITES	[Java, ByteCode, Member, Field]	2519	2120	
29	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, TypeVariable, Bound]	2474	7279	

Graph Density

```
total_number_of_nodes (vertices): 364125
total_number_of_relationships (edges): 1130715
-> total directed graph density: 8.52812094737908e-06
-> total directed graph density in percent: 0.000852812094737908
```